Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
9	(a)	(i)		<u>Enzyme substrate complex</u>	1			1		
		(ii)		Lock and key	1			1		
	(b)	(i)		A. Volume of {oxygen/gas} increases up until {38cm^3/25 minutes} (1) B. (because) {hydrogen peroxide/ substrate} is being {broken down/ used up} (1) C. Volume of {oxygen/ gas} then {remains constant/ stops increasing/ stays the same/ plateaus}/ no gas is produced after {38cm^3/25 minutes} (1) D. (because) <u>all</u> the {hydrogen peroxide/ substrate} has been {broken down/ used up}/ <u>no</u> substrate left (1) IGNORE ref. denature at 25/30min	1 1		1 1	4		1 1
		(ii)		0.4 = 2 marks If incorrect award 1 mark for (7-5)/5 or 2/5		2		2	2	1
		(iii)		• Copper sulfate has bound to (some of the) {catalase/enzyme} / {fewer/some} enzymes are working/ (some) enzymes are not working} (1) • Shape of those active sites has changed / {active site/catalase/enzyme} is denatured (1) • Enzyme substrate complex cannot form / {hydrogen peroxide/substrate} cannot {fit into/ bind with} the {active site/enzyme/catalase} / {No/ fewer} successful collisions between {active site/ catalase/ enzyme} and {hydrogen peroxide/ substrate} (1)		3		3		

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(c)	(i)		<u>Control</u> (experiment/ flask) / {Compare the effect / to show the difference between} of boiled and fresh potato (1)	1			1		1
		(ii)		**Any two (×1) from:** • Temperature (1) • pH (1) • {Mass/weight/ surface area} of potato/ number of potato pieces (1) • {Type/ age} of potato (1) • Concentration of H_2O_2 (1) • Concentration of $CuSO_4$ (1) IGNORE ref. volume/amount/size		2		2		2
		(iii)		Use a { measuring cylinder with a more detailed scale/ burette/ gas syringe} (1)			1	1		1
				Question 9	**5**	**7**	**3**	**15**	**2**	**7**